

Gimbal Mount Assembly

Justification File

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Contents

1. Introduction	1
1.1. Scope	1
1.2. Reference	1
2. Functional Design	2
2.1. Chosen Tolerances and fits	2
2.2. Tolerance Stack-Up Analysis	3
2.3. Preload of NAS-bolt	4
3. Thermal Analysis	5
3.1. Results	5
3.2. Conclusion	7
4. Mechanical Analysis	7
4.1. Bearing Lifetime	7
4.4.2. Spherical Bearing: Liner Stiffness	7
4.5. GMA Static-Structural Analysis	7
5. Conclusion	8
6. Annex	8
6.1. Tolerance Stack Analysis	8
6.2. NAS-bolt torque calculation	9
6.3. Boundary Conditions Thermal Analysis	9
6.4. Boundary Conditions Static Structural Analysis	9
7. Acronym List	9

1. Introduction

1.1. Scope

This document presents the Design Justification File for the Gimbal Mount Assembly (GMA). Its purpose is to explain the rationale behind the selected design and demonstrate that it meets the baseline requirements specified in [RD01]. It lists and describes the justification for:

- Functional Design
- Thermal Design
- Mechanical Design

1.2. Reference

[RD01] Gimbal Mount Assembly - Requirement Consolidation / Version 001

[RD02] Gimbal Mount Assembly - Definition File / Version 001

[RD03] SKF Spherical Plain Bearings and Rod Ends / 2013

[RD04] Bearings, Control System Components, and Associated Hardware Used in the Design and Construction of Aerospace Mechanical Systems and Subsystems / MIL-HDBK-5199 / 1997

[RD05] RBC Aerospace FibriloidCR Series - Cryogenic Rated Plain Bearings: Sphericals, Rod Ends and Journals / 2024

[RD06] SKF Aerospace Solutions / 2020

[RD07] NASA Reference Publication 1228 / Fastener Design Manual / 1990

[RD08] System Requirements Document / TEC-ITA-DOC-2025-01017 / Version 0

2. Functional Design

2.1. Chosen Tolerances and fits

Figure 1 shows a cross-sectional view in which all radial interfaces are color-coded. **Table 1** summarizes the dimensional limits and resulting diametral fit range for each mating outside-diameter/inside-diameter interface. Positive fit values indicate clearance, while negative values indicate interference.

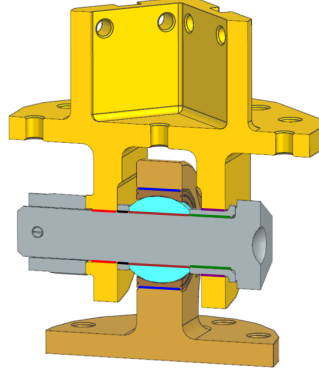


Figure 1: GMA - Radial tolerances color-coded

Table 1: Radial tolerances and fits

Mating parts	Shaft (mm)	Hole (mm)	Resulting Fit (mm)
Bearing & Lug	30.1498–30.1625	30.1570–30.1700	-0.0055 to +0.0202
Bolt & Bearing	15.8369–15.8496	15.8623–15.8750	+0.0127 to +0.0381
Bolt & Clevis	15.8369–15.8496	15.8623–15.8750	+0.0127 to +0.0381
Bolt & Bushing	15.8369–15.8496	15.8623–15.8750	+0.0127 to +0.0381
Bolt & Spacer	15.8369–15.8496	15.8750–15.902	+0.0254 to +0.0651
Bushing & Clevis	19.9800–19.9930	20.0000–20.0210	+0.0070 to +0.0410

The selection of the commercial off-the-shelf (COTS) MS14103-10 spherical bearing and NAS6710DU29 bolt constrains the available bearing and bolt-shank dimensions. The dimensions of the corresponding mating bores were selected to provide the required assembly clearances. The resulting bolt-to-bore clearances are also consistent with established bearing-supplier recommendations, including those provided by *SKF* [RD03].

To simplify assembly and inspection and to minimize the precautions required during integration, a common close-clearance fit is used for most bolt-to-bore interfaces. The exception is the transition fit between the GMA lug-head bore and the spherical-bearing outer ring. During bolt tightening, the clearance at the bolt interfaces allows the components to align and seat axially without radial binding.

The only transition fit in the assembly is between the GMA Lug Head bore and the Spherical Bearing outer ring. This fit intentionally differs from the generic recommendation in the Defense Handbook [RD04], which specifies a nominal 0.001-inch clearance fit.

In contrast to the generic recommendation, *SKF* [RD03] distinguishes between housing fits according to the housing material and the expected loading conditions. For steel/PTFE-fabric spherical bearings subjected to heavier loads or shock loading, *SKF* recommends a tighter housing fit. Applying the recommended *K7*-housing-bore tolerance to the outside-diameter limits of the *MS14103-10* Spherical Bearing results in a calculated fit range of **-0.0180 to +0.0197 mm**.

Other bearing suppliers, including RBC Aerospace [RD05] and NHBB [reference required], recommend a transition fit of approximately -0.005 to +0.020 mm for self-lubricating spherical bearings. The selected GMA dimensions produce a fit range of -0.0055 to +0.0202 mm, which closely matches this recommendation. This fit was selected because the GMA is expected to experience high oscillatory loads during steady-state operation and transient shock loads during ignition, shutdown, and combustion-instability events.

In the event of a interference of **-0.0055 mm** between Spherical Bearing and Lug, the clearance fit is expected to remain between the Bolt and Bearing. As a conservative screening calculation, assuming that the full outer-ring

diametral compression were transferred directly to the bearing bore would leave 0.0072 mm (0.0127 mm - 0.0055 mm) of diametral bolt-to-bearing clearance.

It should also be noted that a tighter housing fit can alter the installed bearing clearance and decrease the bearing friction coefficient at a given temperature, as described qualitatively for self-lubricating spherical bearings by *SKF* [RD06].

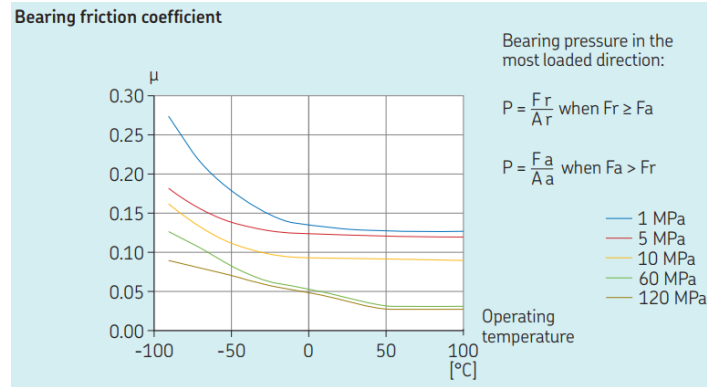


Figure 2: Load and temperature dependend evolution of bearing friction coefficient [RD06]

2.2. Tolerance Stack-Up Analysis

A tolerance stack-up analysis is conducted with the objective to keep tolerances as tight as possible to maintain the design compact, reduce the roll-error (pivot on x-axis) and comply the geometrical constraints given by COTS parts utilized. **Figure 3** shows four cases of tolerance stack-up analysis performed to estimate the minimum and maximum clearance under worst-case considerations (absolute extreme tolerance limits)

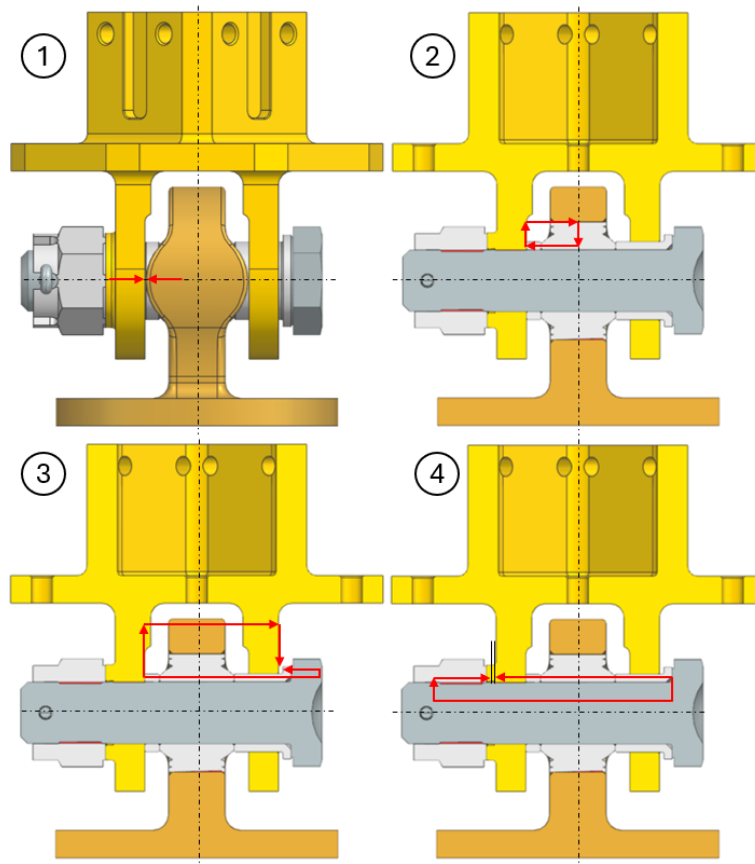


Figure 3: GMA Tolerance Stack-Up

Table 2 presents the related values calculated.

Table 2: Worst-Case Tolerance Analysis

Stack-up	Nominal value (mm)	Max. value (mm)	Min. Value (mm)
1	0.3	0.6	0.1
2	0	0.0854	0.015
3	1.225	1.425	0.904
4	0.699	1.055	0.395

2.3. Preload of NAS-bolt

The Castellated Nut shall be tightened until all components of the axial joint stack are fully seated and unacceptable axial play is eliminated. From this seated condition, the nut shall be advanced in the tightening direction only to the first position at which a castellated slot aligns with the bolt cross-hole, after which the Cotter Pin shall be installed.

Assuming the specified the thread *0.6250-18 UNJF-3A*, the pitch is $p = \frac{1}{18}$ inch = 1.41 mm.

For a castellated nut with six equally spaced slots, the maximum angular advance required for alignment is 60° . The corresponding maximum relative axial thread advance is a pitch stated above. Taking into account that the Castellated Nut *MS9358-016* contains six grooves, the spacing between grooves is 60° . Roughly, for an angle of rotation of 60° with the wrench, the relative thread advance is estimated by:

$$\Delta s = \frac{60^\circ}{360^\circ} * 1.41 \text{ mm} = 0.235 \text{ mm}$$

This value represents the maximum relative nut advance after seating. It shall not be interpreted directly as bolt elongation because the imposed displacement is distributed between bolt extension, compression of the joint stack, and local deformation of the threads and bearing surfaces.

When an adjustment of 0.235 mm is applied in the ANSYS bolt-pretension model, the calculated preload is 63.4 kN. This result includes the modeled stiffness and contact behaviour of the bolt and clamped components but does not account for installation scatter, embedment, relaxation etc.

The analysis with a friction coefficient of 0.2 for the stacked surfaces indicates that this preload significantly changes the lateral load path. Like Figure 4 demonstrates under thrust load (here 22.5 kN), a substantial portion of this load is transferred through friction rather than through the intended bolt, bearing-inner-member, and clevis-hole contact path. Figure 5 shows the consequences of the preload directly mirrored in the Von Mises Stresses. The values are for comparison only as the mesh might be too coarse to assess absolute values.

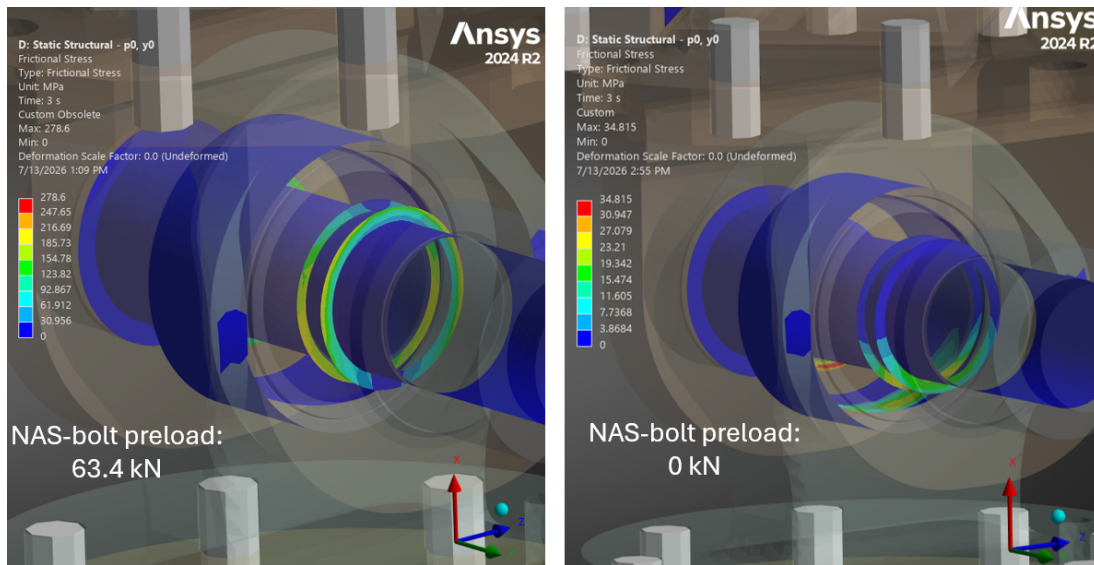


Figure 4: Frictional Stresses with/without preload of NAS-bolt

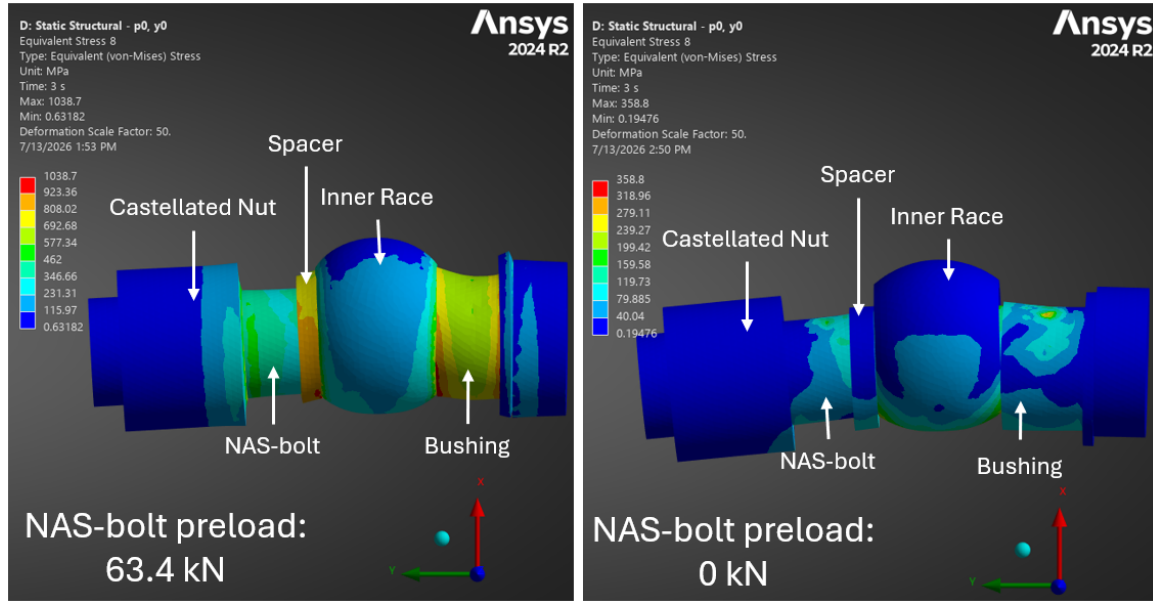


Figure 5: Von Mises Stresses with/without preload of NAS-bolt

In summary, the **minimum torque is to be chosen so that axial play is eliminated**. The maximum torque is estimated so the main engine operational thrust force is not transmitted through the friction as exemplarily shown above. Specifically for the on-Earth demonstrator *Oneiros* a minimum thrust level of 6 kN is maintained through the entire duration as shown by the Guidance Navigation Control simulation (GNC) in Figure 6. Taking into account the *Alternative Torque Formula* by NASA [RD07] for quick bolt torque calculations, a **maximum preload for the NAS-bolt is approximated at 17 Nm**.

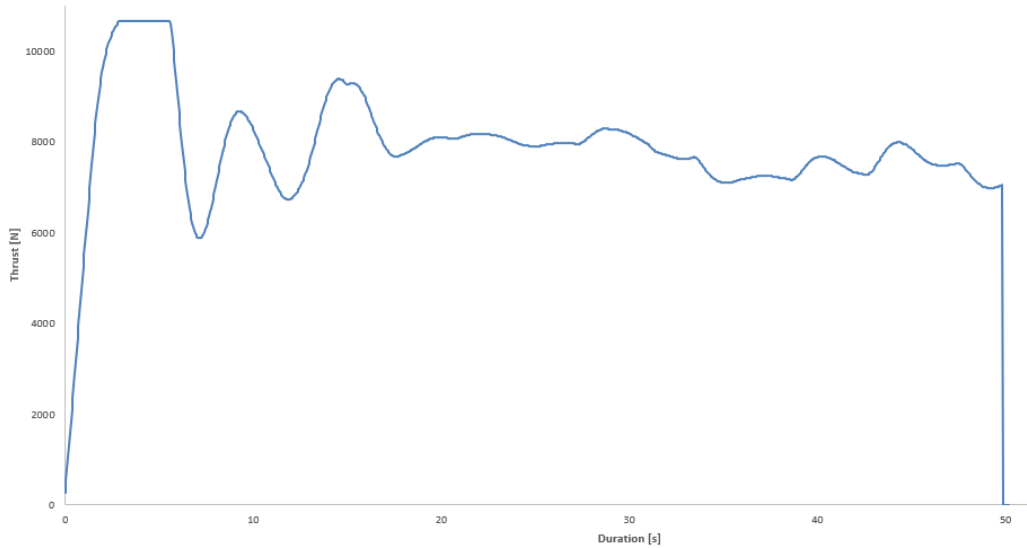


Figure 6: [GNC-simulation - Thrust requirement over Time]

3. Thermal Analysis

3.1. Results

A transient thermal analysis was performed to support the definition of component tolerances specified in the Definition File [RD02] and to reduce the thermal-development risks associated with the main functional requirements of the GMA [RD01] and the *Oneiros* system requirements [RD08].

As part of the preliminary GMA development, a transient thermal model was established. The generic boundary

conditions are summarized in the table below and described in greater detail in the Annex.

Table 3: Generic boundary conditions for the transient thermal analysis

Thermal Boundary	Expression	Comment
Thermal load	90 K	Applied to Thrust Dome flange surface
Ambient Temperature	322.75 K	In accordance with [RD08]]
Flange connections	Bonded	Ideal thermal contact without interface resistance
Heat transfer	Thermal conduction	Radiation and convection neglected

The calculated temperature distributions after mission durations of 60 s for Oneiros and up to 500 s for Terminal Moon are shown in **Figure 7**.

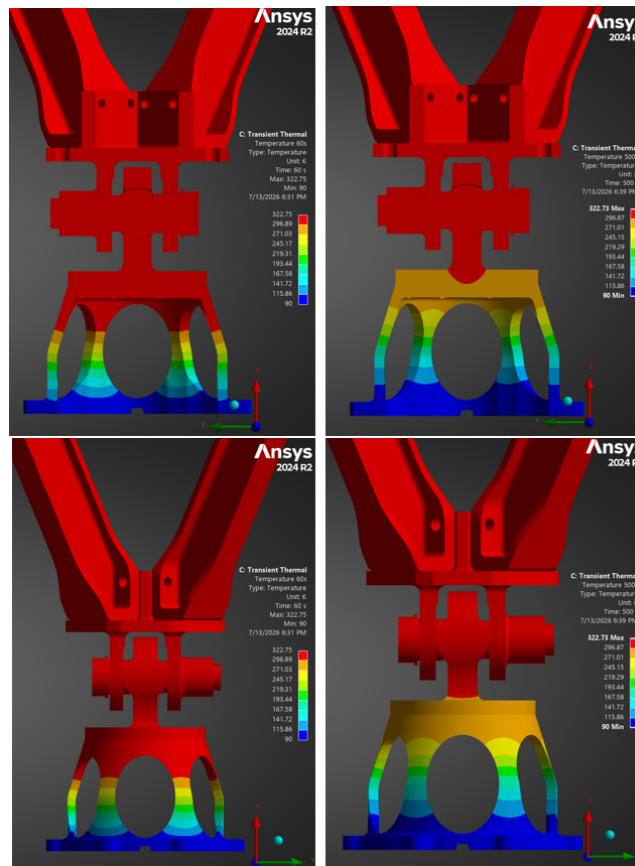


Figure 7: Results of Thermal Transient Analysis for 60 s and 500 s

3.2. Conclusion

A hot-fire duration of 60 s is planned for the *Oneiros* demonstrator [RD08]. The transient thermal-analysis results shown in **Figure7** indicate that, for this duration, no significant temperature gradients are expected in the GMA region containing components subject to relative motion, including the Spherical Bearing and NAS-bolt.

The *Oneiros* demonstration is planned under atmospheric conditions. The present model neglects both convection and radiation and therefore does not represent all heat-transfer mechanisms present during the test. Neglecting convection may be conservative with respect to component cooling. The omission of radiation may likewise be conservative or non-conservative depending on the temperatures and view factors of the surrounding components. For a lunar mission, convection is absent, but radiative heat transfer between the vehicle, the lunar surface, and deep space must be considered in a mission-specific thermal analysis.

A further simplifying assumption is the use of bonded thermal contacts at connected interfaces. This represents ideal, loss-free heat transfer across the flange connections and therefore neglects thermal contact resistance. In the physical assembly, the interface conductance will depend on surface condition, contact pressure, bolt preload, coatings, and interface geometry. For heat transfer from the Thrust Dome into the GMA, the bonded-contact assumption is generally conservative because it maximizes conductive heat transfer across the interfaces.

Despite these simplifications, the analysis provides a useful preliminary estimate for both mission scenarios and identifies the expected thermal propagation path from the Thrust Dome toward the GMA.

The calculated temperature distribution also provides initial guidance for potential future modifications to the Thrust Dome. For example, a reduction in Thrust Dome height would shorten the conductive heat path and could increase the thermal exposure of the GMA, potentially requiring corresponding design changes.

For the current GMA configuration and the short-duration of the *Oneiros* test, the analysis indicates that significant thermal loads are not expected to reach the Lug Head in the vicinity of the Spherical Bearing seat. Consequently, thermal loads are not considered a governing sizing case for the current GMA development stage. This conclusion is limited to the analyzed geometry, boundary conditions, and mission durations and shall be reassessed following significant changes to the Thrust Dome geometry, mission duration, thermal environment, or interface definition.

4. Mechanical Analysis

4.1. Bearing Lifetime

- Show first how size was determined @MIL-HDBK-1599 - 201.143
- Show computed lifetime according SKF
- Discuss results ## 4.2. Bolt, Lug, Clevis
- Preliminary assesment Ro/Ma
- Show simplified assumptions and analytical result
- Show numerical result ## 4.3. Static Structural
- ## 4.4. Preliminary Analysis ### 4.4.1. Bolt, Lug, Clevis

4.4.2. Spherical Bearing: Liner Stiffness

4.5. GMA Static-Structural Analysis

- 0° with Thrust load and Acceleratoin
- Worst case angles (pitch/yaw)
- Discuss results
 - Modelling of liner (boded)

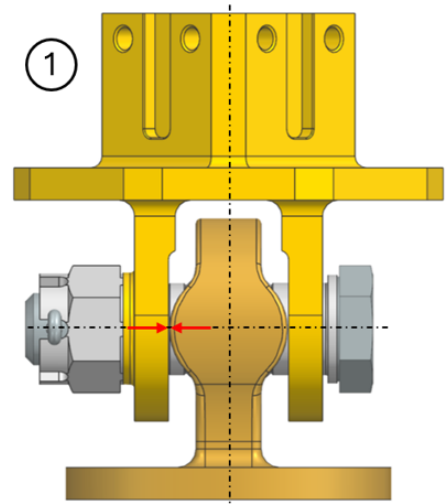
5. Conclusion

6. Annex

6.1. Tolerance Stack Analysis

Total space between bump of lug and clevis inner side (one side only)

Input:					Comment:
Pos.	Direction	Nom. Measu	Upper tol.	Lower tol.	
	R	N _i	T _{io}	T _{iu}	
1	+	26.6	0.1	0	Clevis width
2	+	0	0.015	0.015	Clevis inner parallelism
3	-	26	0	0.2	Lug bump width
4	-	0	0	0.2	Lug bump profile tolerance
5	-	0	0.0704	0	Offset of middle axis of lug
6	-	0	0.1143	0	Wear of bearing liner
7	-	0	0.2	0	Deformation under load
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					



Results:

Arithmetic tolerance analysis:

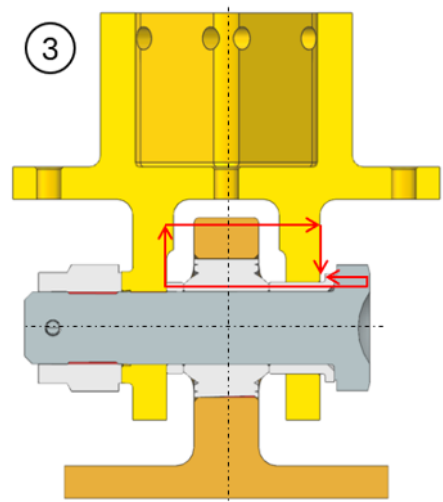
Nom. Value	N ₀ =	0.3
Arithm. Tolerance	T _A =	0.2575 (plus) 0.19985 (minus)

(width=80%)

! [Tolerance stack-up - Lug offset(<../figures/GMA_tolerance_lug offset.png>)] (width=80%)

Offset of Bushing clearance

Input:					Comment:
Pos.	Direction	Nom. Measu	Upper tol.	Lower tol.	
	R	N _i	T _{io}	T _{iu}	
1	-	26.6	0.05	0	Clevis inner dimension
2	+	3.775	0	0.02	Spacer - black
3	+	19.05	0	0.0508	Bearing - green
4	+	14.5	0.05	0.05	Spacer - blue
5	-	7.5	0.1	0.1	Clevis ear thickness
6	-	2	0.05	0.05	Spacer collar - blue
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					



Results:

Arithmetic tolerance analysis:

Nom. Value	N ₀ =	1.225
Arithm. Tolerance	T _A =	0.2 (plus) 0.3208 (minus)

(width=80%)

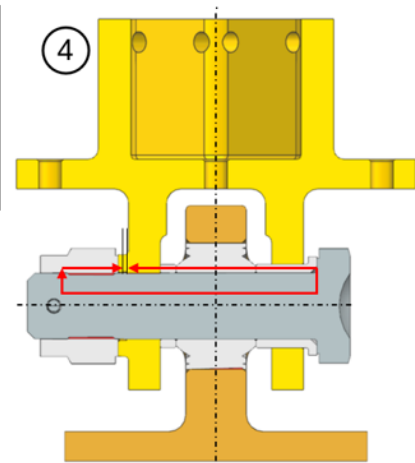
Space between end of thread and begin of grip-diameter of bolt

Input:							Comment:
Pos.	Direction	Nom. Measu	Upper tol.		Lower tol.		
	R	N _i	T _{is}	T _{isR}	T _{is}	T _{isR}	
1	+	46.0248	0.254	0.254	0.254	0.254	Bolthead-Grip NAS6710DU29
2	-	14.5	0.05	0.05	0.05	0.05	Bushing - Blue
3	-	19.05	0	0.051	0.0508	0	Bearing
4	-	3.775	0		0.02	0	Spacer - black
5	-	8	0		0.1	0	Effective width clevis left

Results:

Arithmetic tolerance analysis:

Nom. Value	N ₀ =	0.6998	=SUMME(E10:E18)
Arithm. Tolerance	T _A =	0.3548	=SUMI (plus)
		0.304	=SUMI (minus)



(width=80%)

6.2. NAS-bolt torque calculation

6.3. Boundary Conditions Thermal Analysis

6.4. Boundary Conditions Static Structural Analysis

7. Acronym List

The acronyms used in this document are listed below.

Acronym	Definition
COTS	Commercial Off-The-Shelf
GNC	Guidance Navigation Control
GMA	Gimbal Mount Assembly